Phys 115B HW 2

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Question 1. *More Practice with Spherical Harmonics*
Consider a particle in three dimensions in a state

$$\psi(x,y,z) = A(x+y+2z)e^{-\alpha r}$$

where $\alpha > 0$ is a positive constant, and A a normalization factor.

(a) *Write the spherical harmonics $Y_1^{0,\pm1}$ in terms of x, y, z, and*

$$r = \sqrt{x^2+y^2+z^2}.$$

(b) *Using your result from part (a), show that ψ can be written as a product of a purely r-dependent wave function and a linear combination of spherical harmonics.*

Proof. First, recall that the spherical harmonics are given as follow:

$$Y_\ell^m = \epsilon\sqrt{\frac{2\ell+1}{4\pi}\frac{(\ell-|m|)!}{(\ell+|m|)!}}e^{im\phi}P_\ell^m(\cos(\theta)), \quad \epsilon = \begin{cases}(-1)^m & m \leq 0\\ 1 & m > 0\end{cases} \tag{1}$$

$$P_\ell^m(z) = (1-z^2)^{|m|/2}\left(\frac{d}{dz}\right)^{|m|}\left(\frac{1}{2^\ell \ell!}\left(\frac{d}{dz}\right)^\ell (z^2-1)^\ell\right) \tag{2}$$

Where P_ℓ^m is the associated Legendre's Polynomial.

(a) Given $\ell = 1$ and $m = 0, \pm1$, the associated spherical harmonics (in spherical coordinates) are as follow:

$$\ell = 1,\ m = 0 \implies P_1^0(z) = \frac{1}{2}\frac{d}{dz}(z^2-1) = z \tag{3}$$

$$\implies Y_1^0 = \sqrt{\frac{3}{4\pi}}\cos(\theta) \tag{4}$$

$$\ell = 1,\ m = 1 \implies P_1^1(z) = \sqrt{1-z^2}\frac{d}{dz}\left(\frac{1}{2}\frac{d}{dz}(z^2-1)\right) = \sqrt{1-z^2} \tag{5}$$

$$\implies Y_1^1 = \sqrt{\frac{3}{8\pi}}e^{i\phi}\sqrt{1-\cos^2(\theta)} = \sqrt{\frac{3}{8\pi}}e^{i\phi}\sin(\theta) \tag{6}$$

$$\ell = 1,\ m = -1 \implies P_1^{-1}(z) = \sqrt{1-z^2}\frac{d}{dz}\left(\frac{1}{2}\frac{d}{dz}(z^2-1)\right) = \sqrt{1-z^2} \tag{7}$$

$$\implies Y_1^{-1} = -\sqrt{\frac{3}{8\pi}}e^{-i\phi}\sqrt{1-\cos^2(\theta)} - \sqrt{\frac{3}{8\pi}}e^{-i\phi}\sin(\theta) \tag{8}$$

Then, using the cartesian coordinates $x = r\sin(\theta)\cos(\phi)$, $y = r\sin(\theta)\sin(\phi)$, $z = r\cos(\theta)$, and $r = \sqrt{x^2+y^2+z^2}$, the above becomes the following:

$$Y_1^0 = \sqrt{\frac{3}{4\pi}}\cos(\theta) = \sqrt{\frac{3}{4\pi}}\frac{z}{r} \tag{9}$$

$$Y_1^1 = \sqrt{\frac{3}{8\pi}}(\cos(\phi) + i\sin(\phi))\sin(\theta) = \sqrt{\frac{3}{8\pi}}\frac{x+iy}{r} \tag{10}$$

$$Y_1^{-1} = -\sqrt{\frac{3}{8\pi}}(\cos(\phi) - i\sin(\phi))\sin(\theta) = \sqrt{\frac{3}{8\pi}}\frac{-x+iy}{r} \tag{11}$$

(b) Using the equations (9),(10),(11) from part (a), the linear combinations of $Y_1^{\pm 1}$ provides the following:

$$Y_1^1 + Y_1^{-1} = \sqrt{\frac{3}{8\pi}}\frac{2iy}{r} \implies y = -ir\sqrt{\frac{2\pi}{3}}\left(Y_1^1 + Y_1^{-1}\right) \tag{12}$$

$$Y_1^1 - Y_1^{-1} = \sqrt{\frac{3}{8\pi}}\frac{2x}{r} \implies x = r\sqrt{\frac{2\pi}{3}}\left(Y_1^1 - Y_1^{-1}\right) \tag{13}$$

$$Y_1^0 = \sqrt{\frac{3}{4\pi}}\frac{z}{r} \implies z = \sqrt{2}r\sqrt{\frac{2\pi}{3}}Y_0^1 \tag{14}$$

Hence, the term $x+y+2z$ can be rewritten as:

$$x+y+2z = r\sqrt{\frac{2\pi}{3}}(Y_1^1 - Y_1^{-1}) - ir\sqrt{\frac{2\pi}{3}}(Y_1^1 + Y_1^{-1}) + \sqrt{2}r\sqrt{\frac{2\pi}{3}}Y_1^0 \tag{15}$$

$$= r\sqrt{\frac{2\pi}{3}}((Y_1^1 - T_1^{-1}) - i(Y_1^1 + Y_1^{-1}) + \sqrt{2}Y_1^0) \tag{16}$$

$$= r\sqrt{\frac{2\pi}{3}}((1-i)Y_1^1 - (1+i)Y_1^{-1} + \sqrt{2}Y_1^0) \tag{17}$$

So, the proposed ψ can be given as follow:

$$\psi = A(x+y+2z)e^{-\alpha r} \tag{18}$$

$$= A\sqrt{\frac{2\pi}{3}}((1-i)Y_1^1 - (1+i)Y_1^{-1} + \sqrt{2}Y_1^0) \cdot re^{-\alpha r} \tag{19}$$

Which, this is a product of purely radial-dependent function ($re^{-\alpha r}$), together with linear combinations of the spherical harmonics $((1-i)Y_1^1 - (1+i)Y_1^{-1} + \sqrt{2}Y_1^0)$, up to some scalars.

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Question 2. *Radial Momentum*
In lecture, we defined an operator corresponding to the momentum in the radial direction,

$$p_r \equiv \frac{1}{2}(\hat{r} \cdot p + p \cdot \hat{r}) = -\frac{i\hbar}{2}\left(\frac{1}{r} r \cdot \nabla + \nabla \cdot \frac{r}{r}\right).$$

(a) *Show that $\hat{r} \cdot p$ is not Hermitian, and hence cannot itself be identified with momentum in the radial direction.*

(b) *Using the definition of p_r above, show that*

$$p_r^2 = -\hbar^2 \frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right).$$

Proof. First, recall the definition of these operators:

$$\hat{r} = \frac{x}{r}\hat{x} + \frac{y}{r}\hat{y} + \frac{z}{r}\hat{z} \tag{20}$$

$$\bar{p} = \hat{x}\left(-i\hbar\frac{\partial}{\partial x}\right) + \hat{y}\left(-i\hbar\frac{\partial}{\partial y}\right) + \hat{z}\left(-i\hbar\frac{\partial}{\partial z}\right) \tag{21}$$

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Question 3. *The Size of Hydrogen (Griffiths 4.15, modified, 4.13 in second edition, still modified). Do parts (a)–(c) from the Griffiths problem.*

(a) *Find* $\langle r \rangle$ *and* $\left\langle r^2 \right\rangle$ *for an electron in the ground state of hydrogen. Express your answers in terms of te Bohr radius.*

(b) *Find* $\langle x \rangle$ *and* $\left\langle x^2 \right\rangle$ *for an electron in the ground state of hydrogen.*

(Hint*: This requires no new integration - note that* $r^2 = x^2 + y^2 + z^2$*, and exploit the symmetry of the ground state).*

(c) *Find* $\left\langle x^2 \right\rangle$ *in the state* $n = 2, \ell = 1, m = 1$*.*

*(*Hint: *this staet is* not *symmetrical in* x, y, z*. Use* $x = r\sin(\theta)\cos(\phi)$*.*

(d) *Find* $\langle r \rangle$ *and* $\langle r^2 \rangle$ *for an electron in a circular orbit of hydrogen with arbitrary principal quantum number* n *(corresponds to* $l = n - 1$ *and any allowed* m*).*

(e) *Compute the RMS uncertainty*

$$\sqrt{\langle r^2 \rangle - \langle r \rangle^2}$$

in terms of r *for the electron in part (d). Note that the fractional spread in* r *decreases with increasing* n *(in this sense the system "begins to look classical" for large* n*). How much more volume does a hydrogen atom in the* $n = 100$ *state occupy compared to the hydrogen atom in the ground state. (Hint — you might want to look at Griffiths 4.55, or 4.15 in the second edition.)*

Proof. $\square$

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Question 4. *The Size of Hydrogen continued (Griffiths 4.16, 4.14 in the second edition)*
What is the most probable value of r, in the ground state of hydrogen? (The answer is not zero!)
Hint: First you must figure out the probability that the electron would be found between r and $r + dr$.

Proof. Given the ground state of hydrogen atom as the following:

$$\psi_{100} = \frac{1}{\sqrt{\pi a^3}} e^{-r/a} \tag{22}$$

Then, under this position basis, the expectation value of radius is given as follow:

$$\begin{aligned}
\langle r \rangle = \int_{\mathbb{R}^3} r|\psi|^2 dV &= \int_{r=0}^{\infty} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{r}{\pi a^3} e^{-2r/a} r^2 \sin(\theta) d\phi \; d\theta \; dr && (23) \\
&= \frac{2\pi}{\pi a^3} \left(\int_0^{\infty} r^3 e^{-2r/a} dr \right) \left(\int_0^{\pi} \sin(\theta) d\theta \right) && (24) \\
&= -\frac{4}{a^3} \left(\frac{a}{2} r^3 + \frac{3a^2}{4} r^2 + \frac{3a^3}{4} r + \frac{3a^4}{8} \right) e^{-2r/a} \Big|_0^{\infty} && (25) \\
&= \frac{4}{a^3} \cdot \frac{3a^4}{8} = \frac{3a}{2} && (26)
\end{aligned}$$

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Question 5. *Where is the Electron (Griffiths 4.54, 4.45 in the second edition)*
What is the probability that an electron in the ground state of hydrogen will be found inside the nucleus*?*

(a) *First calculate the* exact answer*, assuming the wave function is correct all the way down to* $r = 0$*. Let* b *be the radius of the nucleus.*

Here is the wave function:

$$\psi_{100}(r, \theta, \phi) = \frac{1}{\sqrt{\pi a^3}} e^{-r/a}$$

(b) *Expand your result as a power series in the small number* $\epsilon := \frac{2b}{a}$*, and show that the lowest-order term is the cubic:* $p \approx \frac{4}{3}\left(\frac{b}{a}\right)^3$*. This should be a suitable approximation, provided that* $b \ll a$ *(which it is).*

(c) *Alternatively, we might assume that* $\psi(r)$ *is essentially constant over the (tiny) volume of the nucleus, so that* $P \approx \frac{4}{3}\pi b^3 |\psi(0)|^2$*. Check that you get the same answer this way.*

(d) *Use* $b \approx 10^{-15}$ *m and* $a \approx 0.5 \cdot 10^{-10}$ *m to get a numerical estimate for* P*. Roughly speaking, this represents the "fraction of its time that the electron spends inside the nucleus".*

Proof.

(a) To calculate the probability P that an electron will be found inside the nucleus (given it's in ground state) is given as follow (assuming the nucleus is of a solid ball with radius b):

$$P = \int_{\|\bar{r}\| \leq b} |\psi_{100}|^2 dV = \int_{r=0}^{b} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{1}{\pi a^3} e^{-2r/a} r^2 \sin(\theta) d\phi d\theta dr \tag{27}$$

$$= \frac{2\pi}{\pi a^3} \left(\int_0^b r^2 e^{-2r/a} dr \right) \left(\int_0^\pi \sin(\theta) d\theta \right) = \frac{4}{a^3} \left(-\frac{a}{2} r^2 - \frac{a^2}{2} r - \frac{a^3}{4} \right) e^{-2r/a} \Big|_0^b \tag{28}$$

$$= 1 - \left(\frac{2b^2}{a^2} + \frac{2b}{a} + 1 \right) e^{-2b/a} \tag{29}$$

(b) Define $\epsilon := \frac{2b}{a}$, then under the case $b \ll a$, one has $\epsilon \approx 0$. So, using the power series expansion, $e^{-2b/a} = e^{-\epsilon}$ has the form $e^{-\epsilon} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \epsilon^n$.

Plugin to the value derived in part (a), one yields the following:

$$P = 1 - \left(\frac{1}{2}\left(\frac{2b}{a}\right)^2 + \frac{2b}{a} + 1\right) e^{-2b/a} = 1 - \left(\frac{\epsilon^2}{2} + \epsilon + 1\right) e^{-\epsilon} \tag{30}$$

$$= 1 - \left(\frac{\epsilon^2}{2} + \epsilon + 1\right) \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \epsilon^n \tag{31}$$

$$= 1 - \left(\sum_{n=0}^{\infty} \frac{(-1)^n}{2(n!)} \epsilon^{n+2} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \epsilon^{n+1} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \epsilon^n\right) \tag{32}$$

$$= 1 - \left(\sum_{n=2}^{\infty} \frac{(-1)^{n-2}}{2((n-2)!)} \epsilon^n + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(n-1)!} \epsilon^n + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \epsilon^n\right) \tag{33}$$

$$= 1 - \left(\sum_{n=2}^{\infty} (-1)^n \left(\frac{1}{2 \cdot (n-2)!} - \frac{1}{(n-1)!} + \frac{1}{n!}\right) \epsilon^n + \epsilon - \epsilon + 1\right) \tag{34}$$

$$= \sum_{n=2}^{\infty} (-1)^{n+1} \left(\frac{1}{2 \cdot (n-2)!} - \frac{1}{(n-1)!} + \frac{1}{n!}\right) \epsilon^n \tag{35}$$

Which, for $n = 2$, the coefficient is $\frac{1}{2 \cdot 0!} - \frac{1}{1!} + \frac{1}{2!} = \frac{1}{2} - 1 + \frac{1}{2} = 0$; nonetheless, for $n = 3$, the coefficient is $\frac{1}{2 \cdot 1!} - \frac{1}{2!} + \frac{1}{3!} = \frac{1}{6}$. This shows that the loest degree term of the power series $P(\epsilon)$ is the cubic.

Finally, plugin the definition $\epsilon := \frac{2b}{a}$, the cubic term becomes:

$$\frac{1}{6}\epsilon^3 = \frac{8}{6}\frac{b^3}{a^3} = \frac{4}{3}\left(\frac{b}{a}\right)^3 \tag{36}$$

(c) Given that $\psi(0) = \frac{1}{\sqrt{\pi a^3}}$, one has the proposed estimate being as follow:

$$P \approx \frac{4}{3}\pi b^3 |\psi(0)|^2 = \frac{4}{3}\pi b^3 \frac{1}{\pi a^3} = \frac{4}{3}\left(\frac{b}{a}\right)^3 \tag{37}$$

Which is exactly as what is proposed in part (b).

(d) Given $b \approx 10^{-15}$ m and $a \approx \frac{1}{2} \cdot 10^{-10}$ m, then one has $\frac{b}{a} \approx 2 \cdot 10^{-5}$. So, one has the following approximation of P:

$$P \approx \frac{4}{3}\left(\frac{b}{a}\right)^3 \approx \frac{32}{3} \cdot 10^{-15} \approx 1.067 \cdot 10^{-16} \tag{38}$$

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